

PROVISIONAL PATENT APPLICATION

METHOD AND SYSTEM FOR RUNTIME GOVERNANCE, COMPLIANCE ATTESTATION, AND CONTINUOUS VERIFICATION OF COMPUTATIONAL TRUST INFRASTRUCTURE

計算信任基礎設施之運行時治理、
合規證明與持續驗證方法與系統

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CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is related to U.S. Patent Application Serial No. [Patent #1] (Trust Source), [Patent #2] (Trust Integrity), [Patent #3] (Trust Evidence), and [Patent #4] (Trust Coordination).

[0002] The present invention provides the Trust Runtime layer (Layer 5) of the Computational Trust Infrastructure. Where Patents #1–#4 address the generation, integrity, packaging, and propagation of trust, the present invention addresses continuous governance over the entire trust infrastructure — ensuring that the system operates within policy, generates compliance attestations, monitors trust health, and enables regulatory framework mapping.

[0003] The boundary between Patent #4 (Trust Coordination) and this invention is precise: **Coordination is horizontal** — governing trust flow between peers. **Runtime is vertical** — governing trust enforcement from top to bottom of the infrastructure. Coordination is a protocol for trust transmission. Runtime is a governance engine for trust assurance.

FIELD OF THE INVENTION

[0004] The present invention relates to systems and methods for continuous runtime governance, automated compliance attestation generation, trust health monitoring, anomaly detection, and regulatory framework mapping for computational trust infrastructure. More specifically, the invention relates to: runtime policy enforcement across trust lifecycles; machine-readable compliance attestation generation; multi-dimensional trust health monitoring with anomaly detection; and pluggable regulatory framework adapters.

BACKGROUND OF THE INVENTION

The Governance Gap in Trust Infrastructure

[0005] Systems that generate, package, and propagate trust artifacts (as described in Patents #1–#4) require continuous governance to ensure correct operation over time. Per-request authorization (controlling who may consume a trust primitive) is insufficient for system-wide governance. The following governance challenges are not addressed by per-request controls:

[0006] **Cross-lifecycle policy enforcement.** Policies that span multiple trust lifecycles — such as "no single entropy source may dominate more than 70% of requests in any 24-hour window" or "all trust artifacts above a certain quality threshold must be blockchain-anchored within 60 seconds" — cannot be evaluated at the per-request level. They require continuous monitoring across the entire trust infrastructure.

[0007] **Compliance attestation.** Regulatory frameworks (EU AI Act, NIST AI RMF, GLI-19, ISO 42001, SOX, HIPAA, FDA 21 CFR Part 11) increasingly require machine-readable compliance evidence for AI systems. Currently, compliance evidence is manually produced audit reports. There is no infrastructure for automatically generating structured compliance attestations that map trust operations to regulatory requirements.

[0008] **Trust health monitoring.** Trust infrastructure may degrade over time: entropy sources may become intermittent, quality scores may drift downward, verification failure rates may increase. Without continuous monitoring, these degradations accumulate silently until a failure occurs — at which point the damage (undetected low-quality trust artifacts already consumed) may be irreversible.

[0009] **Regulatory framework diversity.** Different jurisdictions and industries impose different trust requirements. A gaming operator must satisfy GLI-19. A healthcare AI must satisfy

HIPAA and FDA requirements. An EU-based AI system must satisfy the AI Act. Manually mapping trust operations to each framework is error-prone and does not scale.

[0010] There exists a need for a runtime governance layer that continuously monitors trust infrastructure, enforces cross-lifecycle policies, generates machine-readable compliance attestations, detects anomalies, and adapts to multiple regulatory frameworks through pluggable adapters.

SUMMARY OF THE INVENTION

[0011] The present invention provides:

1. **Runtime Policy Engine:** A system for defining, evaluating, and enforcing trust policies that span multiple trust lifecycles, including trust chain depth limits, entropy source rotation requirements, anchoring frequency requirements, temporal consistency enforcement, and quality score thresholds.
2. **Compliance Attestation Generator:** A system for automatically generating structured, machine-readable compliance attestations that map trust infrastructure operations to specific regulatory framework requirements.
3. **Trust Health Monitor:** A system for continuously computing trust health metrics across the infrastructure, with configurable anomaly detection and alerting.
4. **Regulatory Framework Adapters:** Pluggable modules that map trust lifecycle operations to specific regulatory requirements, enabling automatic compliance reporting for multiple jurisdictions and industries simultaneously.
5. **Trust Audit Interface:** A structured interface for third-party auditors to query, traverse, and verify trust infrastructure operations.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

1. Runtime Policy Engine

[0012] The Runtime Policy Engine evaluates and enforces trust policies that cannot be assessed at the per-request level. Policies are defined as structured rules with the following components:

Policy Component	Description	Example
Scope	What the policy applies to	All trust artifacts, specific consumers, specific verticals
Metric	What is measured	Source concentration, chain depth, anchoring latency, ECS average
Window	Time period for evaluation	Rolling 1 hour, 24 hours, 7 days
Threshold	Acceptable boundary	Source concentration < 70%, chain depth ≤ 5 , anchoring latency < 60s
Action	What happens on violation	Alert, degrade, block, log

[0013] Example policies enforced by the Runtime Policy Engine:

- **Source rotation:** No single entropy source may provide more than a configurable percentage (default: 70%) of entropy contributions in any rolling 24-hour window.

- **Chain depth limit:** No trust chain (DAG path from leaf to root, as constructed by Patent #4) may exceed a configurable maximum depth (default: 5 levels).
- **Anchoring frequency:** All trust artifacts with $ECS \geq$ configurable threshold (default: 800, AA grade) must be anchored to an immutable ledger within a configurable time window (default: 60 seconds).
- **Temporal consistency:** Trust artifact timestamps within a trust chain must be monotonically non-decreasing (no future-dated artifacts, no out-of-order timestamps).
- **Minimum quality floor:** All trust artifacts served to consumers in a designated regulatory context must meet a minimum ECS threshold (e.g., AAA for gaming, AA for enterprise AI).

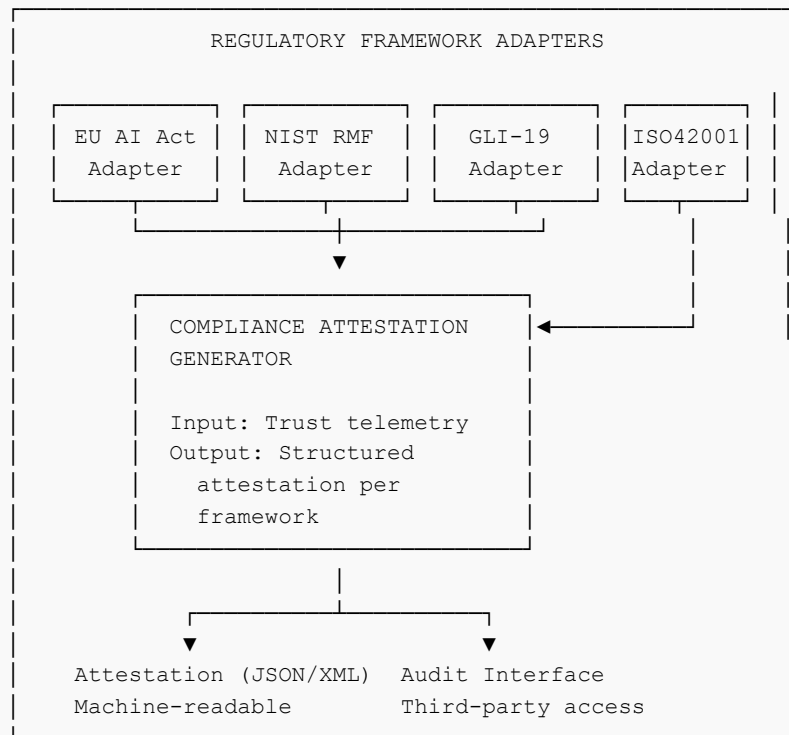
2. Compliance Attestation Generator

[0014] The Compliance Attestation Generator produces structured, machine-readable attestations that map trust infrastructure operations to regulatory framework requirements. Each attestation contains:

- Framework identifier:** The regulatory framework being attested against (e.g., "EU_AI_ACT_2024", "NIST_AI_RMF_1.0", "GLI-19_3.1", "ISO_42001_2023").
- Requirement mapping:** A list of specific requirements from the framework and how the trust infrastructure satisfies each one.
- Evidence references:** Links to specific trust artifacts, trust chains, and policy evaluation records that constitute the compliance evidence.
- Attestation timestamp and signature:** When the attestation was generated and a cryptographic signature from the runtime governance system.
- Coverage period:** The time window covered by the attestation.

f. **Pass/fail status:** Whether all mapped requirements were satisfied during the coverage period.

FIG. 1 — Compliance Attestation Architecture



[0015] Example mapping — EU AI Act Article 14 (Human Oversight): The trust lifecycle audit trail demonstrates that every AI decision backed by a trust artifact is traceable, the entropy sources are identified, the quality score is recorded, and the decision can be independently verified. The compliance attestation maps this capability to Article 14's transparency requirements.

3. Trust Health Monitor

[0016] The Trust Health Monitor continuously computes system-wide trust metrics:

Metric	Measurement	Anomaly Condition
Average ECS	Rolling mean ECS across all issued artifacts	Sustained decline > configurable threshold (e.g., >50 points over 24h)
Source availability	Per-source uptime percentage	Any source < configurable threshold (e.g., <90% over 1h)
Source diversity	Distribution of source types across artifacts	Single source type exceeding concentration threshold
Verification success rate	Percentage of artifacts passing all verification checks	Rate below configurable threshold (e.g., <99.5%)
Anchoring latency	Time between artifact creation and anchor confirmation	Mean latency exceeding configurable threshold
Chain depth distribution	Distribution of trust chain depths (from Patent #4)	Chains exceeding depth policy
Consumption patterns	Per-consumer request rates and patterns	Unusual patterns (bursts, anomalous request volumes)

[0017] When an anomaly is detected, the system generates a structured alert containing: the metric, the observed value, the threshold, the severity level (warning, critical), and recommended actions. Alerts can trigger automated responses (e.g., temporarily increasing the source diversity requirement) or be routed to human operators.

4. Regulatory Framework Adapters

[0018] Regulatory Framework Adapters are pluggable modules that translate between trust infrastructure operations and specific regulatory requirements. Each adapter defines:

- a. **Requirement catalog:** A structured list of the framework's requirements relevant to computational trust.
- b. **Mapping rules:** For each requirement, how trust infrastructure operations satisfy it (which metrics, which artifacts, which policies).
- c. **Evidence collection:** What data must be collected and preserved to demonstrate compliance.
- d. **Reporting format:** The structure and format of compliance reports for the framework.

[0019] The adapter architecture is pluggable: new regulatory frameworks can be supported by adding a new adapter without modifying the core trust infrastructure. Multiple adapters can operate simultaneously, generating compliance attestations for multiple frameworks from the same trust telemetry data.

5. Trust Audit Interface

[0020] The Trust Audit Interface provides structured access for third-party auditors to examine trust infrastructure operations:

- a. **Artifact query:** Retrieve trust artifacts by ID, time range, consumer, quality score range, or source type.
- b. **Chain traversal:** Given any trust artifact, traverse its complete trust chain to the root.
- c. **Policy evaluation history:** Retrieve the history of policy evaluations, violations, and enforcement actions.
- d. **Health metric history:** Retrieve historical trust health metrics for trend analysis.

- e. **Compliance attestation archive:** Retrieve past compliance attestations for any regulatory framework.

[0021] The audit interface does not grant the auditor write access to any trust data. All access is read-only. The auditor can independently verify any trust artifact using the verification methods described in Patent #3, without relying on the audit interface's assertions.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 — Compliance attestation architecture showing regulatory framework adapters, attestation generator, and audit interface.

FIG. 2 — Runtime policy engine showing policy definition, evaluation, and enforcement flow.

FIG. 3 — Trust health monitoring dashboard showing continuous metric computation and anomaly detection.

CLAIMS

What is claimed is:

1. A computer-implemented method for runtime governance of a computational trust infrastructure, comprising:

- (a) receiving trust telemetry data from a trust infrastructure that generates, scores, signs, and distributes trust artifacts;

(b) evaluating the trust telemetry against a plurality of runtime policies, each policy specifying a scope, a metric, a time window, a threshold, and an enforcement action;

(c) detecting policy violations by comparing measured metrics against policy thresholds over rolling time windows; and

(d) executing enforcement actions in response to detected violations, wherein enforcement actions include at least one of: generating an alert, degrading the affected trust service, blocking the affected operation, and recording the violation.

2. The method of claim 1, wherein the runtime policies include at least one of: entropy source concentration limits, trust chain depth limits, immutable ledger anchoring frequency requirements, temporal consistency enforcement, and minimum quality score floors for designated regulatory contexts.

3. A computer-implemented method for generating compliance attestations for a computational trust infrastructure, comprising:

(a) receiving trust telemetry data from a trust infrastructure;

(b) mapping the trust telemetry to requirements of a regulatory framework using a regulatory framework adapter;

(c) generating a structured, machine-readable compliance attestation comprising: a framework identifier, a requirement-by-requirement mapping showing how trust operations satisfy each requirement, references to specific trust artifacts constituting compliance evidence, and a pass/fail status; and

(d) cryptographically signing the compliance attestation.

4. The method of claim 3, wherein the regulatory framework adapter is a pluggable module, and wherein multiple adapters operate simultaneously to generate compliance

attestations for multiple regulatory frameworks from the same trust telemetry data.

5. The method of claim 3, wherein the regulatory framework is selected from the group consisting of: EU AI Act, NIST AI Risk Management Framework, GLI-19, GLI-33, ISO 42001, SOX, HIPAA, FDA 21 CFR Part 11, and FedRAMP.

6. A computer-implemented method for monitoring trust health of a computational trust infrastructure, comprising:

(a) continuously computing a plurality of trust health metrics from trust infrastructure telemetry, the metrics including at least: average trust quality score over time, entropy source availability, source diversity distribution, verification success rate, and anchoring latency;

(b) evaluating each metric against configurable anomaly thresholds; and

(c) generating structured anomaly alerts when metrics exceed thresholds, each alert comprising the metric identifier, observed value, threshold, severity level, and recommended action.

7. The method of claim 6, further comprising automatically adjusting trust infrastructure parameters in response to detected anomalies, including increasing source diversity requirements when source concentration is detected or increasing anchoring frequency when anchoring latency exceeds threshold.

8. A system for runtime governance and compliance of a computational trust infrastructure, comprising:

one or more processors; and

memory storing instructions that, when executed, cause the system to:

(a) continuously receive trust telemetry from a trust infrastructure;

- (b) evaluate the telemetry against runtime policies spanning multiple trust lifecycles;
- (c) generate compliance attestations mapping trust operations to regulatory framework requirements;
- (d) monitor trust health metrics and detect anomalies; and
- (e) provide a read-only audit interface for third-party examination of trust operations, policy history, health metrics, and compliance attestations.

9. The system of claim 8, wherein the compliance attestation generator supports a plurality of regulatory framework adapters operating simultaneously, each adapter mapping trust operations to a different regulatory framework.

10. The system of claim 8, further comprising an automated response engine that adjusts trust infrastructure parameters in response to detected policy violations or health anomalies without requiring human intervention.

ABSTRACT

A method and system for runtime governance, compliance attestation, and continuous verification of computational trust infrastructure. A runtime policy engine evaluates trust telemetry against cross-lifecycle policies (source concentration limits, trust chain depth limits, anchoring frequency requirements, quality score floors) and enforces violations through alerts, degradation, blocking, or logging. A compliance attestation generator automatically produces structured, machine-readable compliance attestations mapping trust operations to regulatory framework requirements (EU AI Act, NIST AI RMF, GLI-19, ISO 42001, SOX, HIPAA) using pluggable regulatory framework adapters. A trust health monitor continuously computes system-wide trust metrics

(average quality scores, source availability, diversity, verification success, anchoring latency) and detects anomalies against configurable thresholds. A trust audit interface provides third-party auditors read-only access to trust artifacts, chain traversals, policy history, health metrics, and compliance attestation archives. The system provides the governance layer (Layer 5) of a computational trust infrastructure, operating vertically across all layers to ensure continuous policy compliance and trust assurance.